

Ramanujan Challenge 2026, Problem 3.1: the 7_2 regulator integral is $4\pi^2/85$

Abstract

Let $A_{7_2}(M, L)$ and the real branch $L = y(M)$ be as in Problem 3.1. We give a computer-assisted proof that

$$\int_{\alpha}^{\beta} \left(\log x \frac{dy}{y} - \log y \frac{dx}{x} \right) = \frac{4\pi^2}{85}.$$

The proof constructs an exact continuous real Riley and enhanced-Ptolemy lift of the full challenge branch. Its left endpoint factors through the $(-1, 2)$ filling and its right endpoint through the $(-1, 1)$ filling. Four exact algebraic shapes at each endpoint give Neumann extended Rogers sums S_{α}, S_{β} . Reciprocal Galois symmetry and an orientation-preserving symmetry of the four-tetrahedron triangulation put twice each extended Bloch class in a totally real field. The regulator values therefore lie on an explicit finite lattice. A 700-bit interval evaluation, with every logarithmic edge and filling equation checked, is much narrower than one lattice spacing and selects

$$S_{\alpha} = -\frac{26\pi^2}{15}, \quad S_{\beta} = -\frac{86\pi^2}{51} \pmod{\pi^2/2}.$$

Zickert's Dehn-invariant formula and the differential of the extended Rogers dilogarithm identify $S_{\beta} - S_{\alpha}$ with the required regulator integral modulo $\pi^2/2$. A coarse positive bound $0 < I < \pi^2/2$ fixes the real lift. A separate signed-coordinate evaluation reproduces the result to about 180 decimal places.

1 The curve and its two representations

Write

$$\eta = \log M \, d \log L - \log L \, d \log M.$$

This is the real A-polynomial Chern–Simons variation form studied by Khoi [2]. The challenge branch runs from $(M, L) = (\alpha, \sqrt{\alpha})$ to (β, β) . The endpoint fields used below are

$$f_{\alpha}(t) = t^{12} - 3t^{11} + 4t^{10} - 5t^9 + 6t^8 - 7t^7 + 7t^6 - 7t^5 + 6t^4 - 5t^3 + 4t^2 - 3t + 1, \quad (1)$$

$$\begin{aligned} f_{\beta}(r) = & r^{16} - 7r^{15} + 22r^{14} - 48r^{13} + 87r^{12} - 133r^{11} + 178r^{10} - 211r^9 + 223r^8 \\ & - 211r^7 + 178r^6 - 133r^5 + 87r^4 - 48r^3 + 22r^2 - 7r + 1. \end{aligned} \quad (2)$$

Take the real roots

$$t = 0.59098942867025644049 \dots, \quad r = 0.40681308133678976238 \dots$$

Exact substitution gives

$$(M_{\alpha}, L_{\alpha}) = (t^2, t), \quad (M_{\beta}, L_{\beta}) = (r, r), \quad A_{7_2}(M_j, L_j) = 0.$$

We identify the two characters independently of the triangulation. For the two-bridge presentation of $7_2 = b(11, 5)$, set

$$a = \begin{pmatrix} s & 1 \\ 0 & s^{-1} \end{pmatrix}, \quad b = \begin{pmatrix} s & 0 \\ -u & s^{-1} \end{pmatrix},$$

$$w = aba^{-1}b^{-1}aba^{-1}b^{-1}ab, \quad \lambda = w^*wa^{-4},$$

where w^* is w read backwards. The knot group relation is $wa = bw$. The longitude λ is the canonical longitude in this presentation. The challenge longitude is its inverse, so its eigenvalue is $L = \lambda_{11}^{-1}$.

The full challenge arc has an exact real lift, not only endpoint lifts. Set $s = M$ and

$$u(M, L) = \frac{(M^2 - 1)(L - 1)}{M^2 + L}. \quad (3)$$

Direct rational-function arithmetic gives

$$(wa - bw)_{12} = \frac{A_{7_2}(M, L)}{M^6(M^2 + L)^5}, \quad A_{7_2}(M, L) \mid \text{num}(L\lambda_{11} - 1) \quad \text{in } \mathbb{Q}[M, L], \quad (4)$$

and the other three relator entries vanish or are multiples of this entry. It also gives

$$\text{tr}([a, b]) - 2 = \frac{(L - 1)(M^2 - 1)^2(L - M^4)}{M^2(M^2 + L)^2}. \quad (5)$$

On the stated branch, $1/4 < M < 1/2$, $1/4 < L < 1$, and $L > M^4$. Thus every denominator in (3)–(5) is nonzero, and (5) never vanishes. Formula (3) therefore defines a continuous real irreducible representation path over the whole challenge arc. Its peripheral eigenvalues are positive and different from 1, so the peripheral holonomies are purely hyperbolic. The displayed Riley form itself is elementary: after triangularizing one meridian, irreducibility lets one normalize the conjugate meridian as shown above. A reducible knot representation has longitude eigenvalue 1 because its diagonal character factors through the abelianization, where the longitude is zero.

Let $R(s, u)$ be the numerator of the $(1, 2)$ entry of $wa - bw$. The diagonal entries vanish identically and the $(2, 1)$ entry has numerator $uR(s, u)$. Exact Euclidean algorithms over the two endpoint fields give linear gcds:

$$\begin{aligned} \gcd_u(R(t^2, u), t\lambda_{11}(t^2, u) - 1) &= u + 2t^{11} - 5t^{10} + 5t^9 - 7t^8 + 9t^7 - 9t^6 \\ &\quad + 9t^5 - 10t^4 + 7t^3 - 6t^2 + 5t - 2, \end{aligned} \quad (6)$$

$$\begin{aligned} \gcd_u(R(r, u), r\lambda_{11}(r, u) - 1) &= u + r^{15} - 7r^{14} + 22r^{13} - 48r^{12} + 87r^{11} \\ &\quad - 133r^{10} + 178r^9 - 211r^8 + 223r^7 - 211r^6 + 178r^5 \\ &\quad - 133r^4 + 87r^3 - 48r^2 + 21r - 5. \end{aligned} \quad (7)$$

Substitution of the two roots for u makes every entry of $wa - bw$ zero. It also gives the exact matrix identities

$$a^{-1}\lambda^{-2} = I \quad (\alpha), \quad a^{-1}\lambda^{-1} = I \quad (\beta).$$

The two gcd roots are exactly the specializations of (3). Thus the left representation factors through the SnapPy filling $(-1, 2)$ and the right representation through $(-1, 1)$. The identity $L\lambda_{11} = 1$ is also the exact basis bridge: the enhanced-Ptolemy variable L is the challenge and SnapPy standard-longitude eigenvalue, while λ is the canonical longitude in the displayed two-bridge presentation.

2 Exact enhanced Ptolemy points

Use SnapPy's four-tetrahedron triangulation of the 7_2 exterior. Zickert's enhanced Ptolemy variety parametrizes generically decorated representations with prescribed meridian and longitude eigenvalues [5]. After setting $c_{1100,0} = 1$, writing $m = M^{-1}$, and eliminating identified coordinates, the enhanced ideal is the following four-equation system in b, c, e :

$$\begin{aligned} -Mb - mc - be &= 0, \\ -Lmb^2 - m - e &= 0, \\ Lm^4b + Lmbe - c &= 0, \\ Mc^2 - Me^2 + c &= 0. \end{aligned} \tag{8}$$

This system lifts the whole real arc on one fixed triangulation. On $A_{7_2}(M, L) = 0$, put

$$b^2 = \frac{(1 - M^2)(L - M^4)}{LM^2(M^2 + L)}, \quad e = -\frac{Lb^2 + 1}{M}, \quad c = b(Lb^2 + 1 - M^2). \tag{9}$$

The certificate uses an explicit rational choice of $b(M, L)$ and verifies all four equations as rational-function identities modulo A_{7_2} . In particular, it verifies that the difference between its b^2 and the displayed expression for b^2 in (9) is

$$\frac{(M^6 + L)A_{7_2}(M, L)}{L^2(M^2 - 1)^2M^6(M^2 + 1)^2(M^2 + L)^4}.$$

The displayed value of b^2 is positive on the challenge arc. The endpoint solutions both have $b < 0$, so choose the continuous negative branch. Then

$$Lb^2 + 1 - M^2 = \frac{L(1 - M^2)(1 + M^2)}{M^2(M^2 + L)} > 0,$$

and hence b, c, e stay negative and nonzero. Each of the 24 signed coordinates is a fixed sign times a monomial in M, L times one of $1, b, c, e$. Thus none vanishes, the four shapes avoid 0 and 1, and their logarithms can be continued from the left endpoint to the right endpoint without changing branches.

Over $\mathbb{Q}(t)$ at $(M, L) = (t^2, t)$ and over $\mathbb{Q}(r)$ at $(M, L) = (r, r)$, the reduced Groebner basis is linear in b, c, e . The certificate solves it exactly, expands all 24 signed Ptolemy coordinates, and substitutes them back into the unreduced enhanced Ptolemy ideal. Every residual is the zero element of the relevant number field.

Let $z_{j,0}, \dots, z_{j,3}$ be the four primary cross-ratios obtained from those signed coordinates. Direct multiplication in the number fields gives every edge product and the filled-cusp product as 1 exactly. None of the shapes is 0 or 1. The Ptolemy point therefore gives a nondegenerate decorated representation with the stated peripheral eigenvalues. The certificate also checks that its three base coordinates equal the specializations of the rational full-curve coordinates used in (9).

These endpoint Ptolemy representations are the endpoint Riley representations of Section 1. Indeed, $L_j \neq 1$, whereas the longitude eigenvalue of a reducible knot-group representation is 1, so the Ptolemy representation is irreducible. Any irreducible representation of the two-bridge group can be put in the displayed Riley form. The linear gcds (6) and (7) leave one character over each peripheral point. This proves equality of the characters, rather than only equality of their peripheral eigenvalues.

3 Reciprocal symmetry and a finite regulator lattice

Both endpoint polynomials are reciprocal. Let $\tau_\alpha(t) = t^{-1}$ and $\tau_\beta(r) = r^{-1}$. Their fixed fields are generated by $t + t^{-1}$ and $r + r^{-1}$, with defining polynomials

$$g_\alpha(X) = X^6 - 3X^5 - 2X^4 + 10X^3 - X^2 - 7X + 1, \quad (10)$$

$$g_\beta(X) = X^8 - 7X^7 + 14X^6 + X^5 - 25X^4 + 9X^3 + 12X^2 - 3X - 1. \quad (11)$$

Exact real-root isolation gives signatures $(6, 0)$ and $(8, 0)$ and discriminants $5^3 \cdot 3881$ and 17^7 . In particular, both fixed fields are totally real.

The reciprocal action on the exact shapes is the same at both endpoints:

$$\tau(z_0) = 1 - z_2^{-1}, \quad \tau(z_1) = z_1, \quad \tau(z_2) = (1 - z_0)^{-1}, \quad \tau(z_3) = z_3. \quad (12)$$

The four-tetrahedron triangulation has a nonidentity orientation-preserving automorphism

$$0 \mapsto (2; (2, 1, 3, 0)), \quad 1 \mapsto (1; (3, 2, 1, 0)), \quad 2 \mapsto (0; (3, 1, 0, 2)), \quad 3 \mapsto (3; (1, 0, 3, 2)). \quad (13)$$

Here the tuple after the semicolon is the vertex permutation. All four permutations are even. The induced cusp map is $-I$, so the automorphism extends over both fillings. Its shape transformations are exactly (12).

Let E_j be the endpoint field. The exact Riley lift is an $\mathrm{SL}_2(E_j)$ representation of the closed filling, so it defines

$$x_j \in \widehat{\mathcal{B}}_{\mathrm{SL}}(E_j) \simeq K_3^{\mathrm{ind}}(E_j).$$

Write $\bar{x}_j$ for its image in the PSL extended Bloch group. These two groups and their periods must be kept separate: the SL regulator is naturally valued modulo $4\pi^2$ [6, Section 2.2.1], and the PSL Cheeger–Chern–Simons regulator modulo π^2 [3, Proposition 2.5 and Theorem 2.6]. The certified evaluator uses the further quotient modulo $\pi^2/2$ [4, Theorem 3.4]. All maps here are homomorphisms, so an order bound for x_j also bounds the evaluator’s value on $\bar{x}_j$.

The exact shape identities above say that $\tau_j \rho_j$ is conjugate in PSL_2 to $\rho_j \circ \phi_*$, where ϕ is the automorphism (13). Nondegenerate shape data determine the decorated PSL representation. The map ϕ preserves orientation, and its cusp map is $-I$, so it extends over each filling and preserves the closed fundamental class. The two resulting SL lifts can differ only by a $\mathbb{Z}/2$ cocycle. Zickert’s two-torsion result for such a change of lift therefore gives [6, Theorem 6.15]

$$2\tau_j(x_j) = 2x_j.$$

Galois descent [6, Corollaries 7.14–7.15] places $2x_j$ in $K_3^{\mathrm{ind}}(F_j)$, where F_j is the corresponding totally real fixed field.

Borel’s rank theorem gives $\mathrm{rank} K_3^{\mathrm{ind}}(F) = r_2(F)$ [1]. Hence $K_3^{\mathrm{ind}}(F_j)$ is finite because F_j is totally real. For a number field F , its torsion subgroup is cyclic of order

$$w(F) = 2 \prod_p p^{\nu_p}, \quad \nu_p = \max\{\nu : \zeta_{p^\nu} + \zeta_{p^\nu}^{-1} \in F\} \quad (14)$$

[6, Section 1]. We only need degree-based upper bounds. If F is totally real of degree d and contains $\zeta_{p^\nu} + \zeta_{p^\nu}^{-1}$, then $\varphi(p^\nu)/2$ divides d when $p^\nu > 2$. Hence

$$w(F_\alpha) \mid 2 \cdot 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 = 65520, \quad w(F_\beta) \mid 2 \cdot 2^5 \cdot 3 \cdot 5 \cdot 17 = 16320. \quad (15)$$

Thus x_α and x_β have orders dividing 131040 and 32640. Their regulator difference, after projection to $\mathbb{C}/(\pi^2/2)\mathbb{Z}$, has order dividing

$$N = \text{lcm}(131040, 32640) = 8910720.$$

After division by π^2 , it lies on the lattice

$$\frac{1}{2N}\mathbb{Z}/\frac{1}{2}\mathbb{Z} = \frac{1}{17821440}\mathbb{Z}/\frac{1}{2}\mathbb{Z}. \quad (16)$$

4 Certified selection of the regulator value

For each endpoint, the exact Ptolemy coordinates determine logarithmic coordinates w_0, w_1, w_2 on every tetrahedron. The certificate multiplies these logarithms by the full 5×12 PGL gluing matrix. All four edge rows and the row `filling_0_0` contain zero as complex intervals. The largest row diameter is 2.24×10^{-207} at α and 6.28×10^{-206} at β . Because all peripheral eigenvalues are positive, the filling relations have no hidden $2\pi i$ term:

$$-\log M_\alpha + 2\log L_\alpha = 0, \quad -\log M_\beta + \log L_\beta = 0.$$

Thus the closed strong flattenings are the endpoint values of the continuous exterior Ptolemy regulator from Section 2.

The extended Rogers sum is evaluated using 700-bit complex intervals and Neumann’s 6-torsion correction for a triangulation without a vertex ordering [3, Section 13 and Proposition 13.1]. The lifted flattenings satisfy Yoon’s edge and Dehn-filling conditions; Theorem 3.1 and the parity argument in the proof of Theorem 3.4 identify the sum with the filled complex volume modulo $\pi^2/2$ [4]. The correction coefficient is exactly -6 at both endpoints, so it contributes $-6(\pi^2/6) = -\pi^2$, which is zero in the stated quotient and cancels in the difference. In the evaluator’s $i(\text{Vol} + i \text{CS})$ convention, it returns

$$\frac{S_\alpha}{\pi^2} = -1.733333333333333333333333 \dots, \quad (17)$$

$$\frac{S_\beta}{\pi^2} = -1.686274509803921568627451\dots \quad (18)$$

The interval for $(S_\alpha - S_\beta)/\pi^2 + 4/85$ contains zero. Its real diameter is less than 2.90×10^{-207} and its imaginary diameter is less than 3.87×10^{-207} . The lattice spacing in (16) is about 5.61×10^{-8} . Since $4/85$ is itself a lattice point ($8N/85$ is an integer), the interval contains exactly one possible regulator value. Thus

$$S_\beta - S_\alpha = \frac{4\pi^2}{85} \pmod{\pi^2/2}. \quad (19)$$

The same lattice argument selects the representatives $S_\alpha = -26\pi^2/15$ and $S_\beta = -86\pi^2/51$ modulo $\pi^2/2$.

There is a separate numerical path in the certificate. It expands the actual signed Ptolemy coordinates, uses SnapPy’s public cross-ratio and flattening code, and multiplies the logarithmic gluing matrix explicitly. This path uses $\text{Vol} + i \text{CS}$ rather than $i(\text{Vol} + i \text{CS})$ and returns

$$\frac{\text{CS}_\alpha}{\pi^2} = \frac{1}{15}, \quad \frac{\text{CS}_\beta}{\pi^2} = \frac{1}{51} \pmod{1/6},$$

so $\text{CS}_\alpha - \text{CS}_\beta = 4\pi^2/85$ to about 180 decimal places. This second calculation is not used for the lattice proof.

5 From the regulator to the challenge integral

It remains to fix the differential and the period. Let $S(M, L)$ be the continuous extended Rogers sum on the Ptolemy lift (9), with logarithms continued from the left endpoint. The unordered-triangulation correction is constant, so it has zero differential. For $n = 2$, Zickert's Dehn-invariant formula gives $-2M \wedge L$ in a peripheral basis of intersection $+1$ [5, Theorem 1.10]. The certificate's M, L are the challenge and SnapPy standard-basis eigenvalues by the exact identity $L\lambda_{11} = 1$ from Section 1. The standard knot meridian-longitude basis has intersection -1 [5, Section 3.1], so in this basis

$$\nu = 2M \wedge L. \quad (20)$$

As a direct sign check, the certificate forms $z_i \wedge (1 - z_i)$ from the exponent vectors of all 24 signed-coordinate monomials. Every wedge involving b, c, e cancels, and the sole surviving coefficient is $+2$ on $M \wedge L$. For Neumann's extended Rogers function,

$$2 dR(z) = \log z d \log(1 - z) - \log(1 - z) d \log z.$$

Applying this identity to the formal shape sum and using (20) gives

$$dS = \log M d \log L - \log L d \log M = \eta. \quad (21)$$

The continuous Ptolemy lift connects the exact endpoint flattenings, so integrating (21) and using (19) gives

$$I := \int_{\alpha}^{\beta} \eta = \frac{4\pi^2}{85} \pmod{\pi^2/2}. \quad (22)$$

The stated branch fixes the lift without numerical quadrature. Put $X = -\log x$ and $Y = -\log y$. Since $x, y \in (0, 1)$, $dx > 0$, and $dy < 0$,

$$\eta = X dY - Y dX > 0,$$

so $I > 0$. Exact root isolation gives

$$\frac{1}{4} < \alpha < \frac{3}{8} < \beta < \frac{1}{2}, \quad \frac{1}{4} < y(\beta) < \frac{1}{2} < y(\alpha) < 1.$$

It follows that

$$\begin{aligned} I &< \log 4((Y_{\beta} - Y_{\alpha}) + (X_{\alpha} - X_{\beta})) \\ &= \log 4 \log \left(\frac{y(\alpha)}{y(\beta)} \frac{\beta}{\alpha} \right) < \log 4 \log 8 < \frac{9}{2} < \frac{\pi^2}{2}. \end{aligned}$$

Both I and $4\pi^2/85$ lie strictly between 0 and $\pi^2/2$. The congruence (22) therefore has one possible real lift, proving

$$\boxed{\int_{\alpha}^{\beta} \left(\log x \frac{dy}{y} - \log y \frac{dx}{x} \right) = \frac{4\pi^2}{85}.$$

6 Reproducibility

The proof certificate is

`numerics/3.1-riley-ptolemy-cs.py.`

It requires SageMath 10.9 and SnapPy 3.3.2 in Sage’s Python. From the repository root:

```
python3 numerics/3.1-riley-ptolemy-cs.py
python3 numerics/3.1-knot72-integral_verify.py
python3 numerics/3.1-anchor-test.py
```

The main command is the proof certificate. It rejects optimized Python and SnapPy versions other than 3.3.2, then exits zero only after the exact full-curve Riley and Ptolemy lift, endpoint filling, reciprocal-symmetry, fixed-field, logarithmic-gluing, interval-lattice, and independent signed-coordinate checks pass. The other commands are independent checks of the real integral and its neighboring Seifert-anchor segments; they are not inputs to the proof.

References

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